COMMUTATIVITY OF THE Q-ONSAGER ELEMENTS XI_1 AND XI_2 IF AND ONLY IF Q^6 IS NOT 1

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Editorial note. The counted claim is the author Proposition lem:RS2 at source lines 2199-2206, and its proof is the proof environment that immediately follows it at source lines 2208-2398; no other proof in the paper is used for it.

Editorial note. The motivation prose at source lines 2099-2104 points to the alternative presentation Theorem prop:altpres, which is not part of this material; it is omitted because no step of the counted proof depends on it, not because any author argument was dropped.

Editorial note. Definition def:qons carries an author citation in its optional argument; the bibliography fragment sources/sigma-2011-069-askey-wilson/qonsager-commutativity-bibliography.tex reproduces that cited entry verbatim from the source bibliography.

Allowed standard prerequisites (editorial).

- q-integer arithmetic in the standing field. The field F and the nonzero q in F with q^4 not 1 are as fixed at source lines 240-244, and the q-integer [n]_q = (q^n - q^-n)/(q - q^-1) satisfies [3]_q = 0 exactly when q^6 = 1.. Splits the two directions of the counted claim at source lines 2210-2211 and 2324-2326.
- Free product of two cyclic groups of order three. For G = H * K with H and K cyclic of order three and generators h and k, every element of G has a unique reduced alternating normal form, and the elements of G form a basis of the group algebra F G.. Used at source lines 2343-2357 to define the module and at 2382-2392 to certify that the displayed element of F G is nonzero.
- Associative expansion identities. The commutator identity of lem:gen(i)-(ii) is an expansion identity valid in any associative algebra, and a relation of the form X^3 Y - Y X^3 = c (X Y - Y X) makes the element X^3 - c X commute with Y, hence central in O.. Used at source lines 2330-2338 for the degenerate case q^6 = 1.

Definition 0.1 ([1, Definition 3.9])**.** Let $\mathcal{O} = \mathcal{O}_q$ denote the $\mathbb{F}$-algebra defined by generators X, Y and relations

$$X^3Y - [3]_q X^2YX + [3]_q XYX^2 - YX^3 = -\big(q^2 - q^{-2}\big)^2 (XY - YX), \tag{0.1}$$

$$Y^3X - [3]_q Y^2XY + [3]_q YXY^2 - XY^3 = -\big(q^2 - q^{-2}\big)^2 (YX - XY). \tag{0.2}$$

We call $\mathcal{O}$ the *q-Onsager algebra*.

Definition 0.2. Let ξ_1, ξ_2 denote the following elements of $\mathcal{O}$:

$$\xi_1 = XY - YX,$$
$$\xi_2 = X^2Y^2 - Y^2X^2 + \big(q^2 + q^{-2}\big)(YXYX - XYXY).$$

Referring to Definition 0.2, we are going to show that ξ_1, ξ_2 commute if and only if $q^6 \neq 1$. We will use the following results, which apply to any $\mathbb{F}$-algebra.

Lemma 0.3. *Let x, y denote elements in any $\mathbb{F}$-algebra, and consider the commutator*

$$\big[xy - yx,\, x^2y^2 - y^2x^2 + \big(q^2 + q^{-2}\big)(yxyx - xyxy)\big]. \tag{0.3}$$

(*i*) *The element* (0.3) *is equal to*

$$xyx^2y^2 - x^2y^2xy + yxy^2x^2 - y^2x^2yx + x^2y^3x - xy^3x^2 + y^2x^3y - yx^3y^2$$
$$- \big(q^2 + q^{-2}\big)\big(xyxy^2x - xy^2xyx + yxyx^2y - yx^2yxy\big).$$

(*ii*) *The element* (0.3) *times* $[3]_q$ *is equal to*

$$[y,[y,[x,[x,[x,y]\,]_q]_{q^{-1}}]_q]_{q^{-1}} + [x,[x,[y,[y,[y,x]\,]_q]_{q^{-1}}]_q]_{q^{-1}}, \tag{0.4}$$

where $[u,v]_\epsilon$ *means* $\epsilon uv - \epsilon^{-1}vu$.

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Proof. (i) Expand (0.3) and simplify the result. (ii) Expand (0.4) and compare it with the expression in (i) above. □

Proposition 0.4. *Referring to Definition* 0.2*, the elements* ξ_1*,* ξ_2 *commute if and only if* $q^6 \neq 1$*.*

Proof. First assume $q^6 \neq 1$, so that $[3]_q$ is nonzero. Applying Lemma 0.3(ii) to the elements $x = X$ and $y = Y$ in the algebra $\mathcal{O}$, we find that $[\xi_1, \xi_2]$ times $[3]_q$ is equal to

$$[Y,[Y,[X,[X,[X,Y]\,]_q]_{q^{-1}}]_q]_{q^{-1}} + [X,[X,[Y,[Y,[Y,X]\,]_q]_{q^{-1}}]_q]_{q^{-1}}. \tag{0.5}$$

We show that the element (0.5) is zero. Observe

$$\begin{aligned}[X,[X,[X,Y]\,]_q]_{q^{-1}} &= X^3Y - [3]_q X^2YX + [3]_q XYX^2 - YX^3 \\ &= -\big(q^2 - q^{-2}\big)^2 [X,Y] = \big(q^2 - q^{-2}\big)^2 [Y,X].\end{aligned}$$

Similarly

$$[Y,[Y,[Y,X]\,]_q]_{q^{-1}} = \big(q^2 - q^{-2}\big)^2 [X,Y].$$

Therefore

$$[Y,[Y,[X,[X,[X,Y]\,]_q]_{q^{-1}}]_q]_{q^{-1}} = \big(q^2 - q^{-2}\big)^2 [Y,[Y,[Y,X]\,]_q]_{q^{-1}} = (q^2 - q^{-2})^4 [X,Y].$$

Similarly

$$[X,[X,[Y,[Y,[Y,X]\,]_q]_{q^{-1}}]_q]_{q^{-1}} = \big(q^2 - q^{-2}\big)^4 [Y,X].$$

By these comments the element (0.5) is zero, and therefore ξ_1, ξ_2 commute.

Next assume $q^6 = 1$, so that $[3]_q = 0$ and $(q^2 - q^{-2})^2 = -3$. In this case the relations (0.1), (0.2) become

$$X^3Y - YX^3 = 3(XY - YX), \qquad Y^3X - XY^3 = 3(YX - XY).$$

In the above line the relation on the left (resp. right) asserts that $X^3 - 3X$ (resp. $Y^3 - 3Y$) commutes with Y (resp. X) and is therefore central in $\mathcal{O}$. We show that ξ_1, ξ_2 do not commute by displaying an $\mathcal{O}$-module on which $[\xi_1, \xi_2]$ is nonzero. Define a group G to be the free product $H * K$ where each of H, K is a cylic group of order 3. Let h (resp. k) denote a generator for H (resp. K). Let $\mathbb{F}G$ denote the group $\mathbb{F}$-algebra. We give $\mathbb{F}G$ an $\mathcal{O}$-module structure. To do this we specify the action of X, Y on the basis G of $\mathbb{F}G$. For the identity $1 \in G$ let $X.1 = h$ and $Y.1 = k$. For $1 \neq g \in G$ write $g = g_1 g_2 \cdots g_n$ such that for $1 \leq i \leq n$, $g_i \in \{h, h^{-1}\}$ for one parity of i and $g_i \in \{k, k^{-1}\}$ for the other parity of i. Let X, Y act on g as follows:

Case	$g_1 = h$	$g_1 = h^{-1}$	$g_1 = k$	$g_1 = k^{-1}$
$X.g$	hg	$3h^{-1}g$	hg	hg
$Y.g$	kg	kg	kg	$3k^{-1}g$

We have now specified the actions of X, Y on G. These actions give $\mathbb{F}G$ an $\mathcal{O}$-module structure on which $X^3 = 3X$ and $Y^3 = 3Y$. For the $\mathcal{O}$-module $\mathbb{F}G$ we now apply $[\xi_1, \xi_2]$ to the vector 1. By Lemma 0.3(i) and the construction, the element $[\xi_1, \xi_2]$ sends 1 to

$$\begin{aligned}&hkh^{-1}k^{-1} - h^{-1}k^{-1}hk + khk^{-1}h^{-1} - k^{-1}h^{-1}kh + 3h^{-1}kh - 3hkh^{-1} + 3k^{-1}hk - 3khk^{-1} \\ &\quad + hkhk^{-1}h - hk^{-1}hkh + khkh^{-1}k - kh^{-1}khk.\end{aligned}$$

The above element of $\mathbb{F}G$ is nonzero. Therefore $[\xi_1, \xi_2] \neq 0$ so ξ_1, ξ_2 do not commute. □

References

[1] Terwilliger P., Two relations that generalize the q-Serre relations and the Dolan–Grady relations, in Physics and Combinatorics 1999 (Nagoya), World Sci. Publ., River Edge, NJ, 2001, 377–398, math.QA/0307016.